

Ability of a high-total antioxidant capacity diet to increase stool weight and bowel antioxidant status in human subjects.

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There is limited knowledge about the possible effect of unabsorbed dietary antioxidants that reach the large intestine on bowel habits. The aim of the present study was to investigate whether a dietary recommendation directed to increase diet total antioxidant capacity (TAC) is able to affect gut function in human subjects. In this cross-over intervention, nineteen subjects followed a high-TAC (HT) and a low-TAC (LT) diet for 2 weeks, which were comparable for energy, macronutrient, total dietary fibre and alcohol contents. At the end of each intervention period, the 48 h stool output was recorded. In the faecal samples obtained from a subset of nine subjects, moisture, pH, ammonia content, Lactobacillus and Bifidobacterium counts, faecal water antioxidants and genotoxicity were measured. A 3 d weighed food record was used to assess the diet composition during HT and LT diet intake. Significant increases in the intake of TAC, vitamins E and C and phenolic compounds were observed during the HT diet intake. The higher intake of antioxidants led to increased 48 h stool output (324 (SD 38) g in HT v. 218 (SD 22) g in LT), and to higher TAC and total phenolic concentrations in faecal water. No significant variation in the other measured parameters was observed between the diets. In conclusion, a diet selected to raise the intake of dietary antioxidants is able to increase stool bulk and antioxidant content of faeces.